

V. DISCUSSION

A. Applicability to Different Multi-Beam LEO Constellations

The proposed EABR framework is designed for general multi-beam LEO satellite systems and is not limited to a specific constellation. Although the evaluation in this work adopts a OneWeb-like constellation as the simulation case, the core idea of EABR depends on predictable satellite motion, beam footprints, beam-level EPFD contributions, and coverage overlap among neighboring LEO satellites. These properties are common in many multi-beam LEO systems.

Therefore, as long as the satellite geometry and beam coverage can be estimated in advance, the time-slot beam role record and candidate beam relation graphs can be constructed for different constellation configurations. The proposed framework can then identify the critical satellites and helper satellites according to the current system state. In this sense, EABR is applicable to different multi-beam LEO constellations, while the OneWeb-like setting is used as a representative case for evaluation.

B. Helper Availability and Resource-Limited Recovery

The performance of helper-based service recovery depends on the availability of neighboring helper satellites and helper beams. In practical systems, helper availability is affected by constellation-level resource conditions, such as satellite density, orbital spacing, beam footprint overlap, EPFD limit strictness, and the remaining transmit power of neighboring satellites. Therefore, insufficient helper resources should be regarded as a system-level resource limitation rather than a failure of the reassociation framework.

When helper satellite resources are limited, the recovery performance of EABR is bounded by the available helper coverage, helper beam capacity, and helper satellite transmit power. Nevertheless, as long as partial helper coverage or helper power is available, EABR can still exploit these resources for service recovery and provide additional satisfaction improvement over conventional interference mitigation approaches. The framework also maintains the EPFD constraint and power constraints during the reassociation process, ensuring that service recovery does not introduce additional interference violation.

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